

EDA LOGISTIC REGRESSION WITH EDA

CODE and OUTPUT:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score

df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition.csv")

df.head()
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	..
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	..
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	..
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	..
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	..
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	..

5 rows × 35 columns

```
df.shape
```

```
df.info()
```

```
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Age                   1470 non-null  int64
 1   Attrition             1470 non-null  object
 2   BusinessTravel        1470 non-null  object
 3   DailyRate             1470 non-null  int64
 4   Department            1470 non-null  object
 5   DistanceFromHome      1470 non-null  int64
 6   Education             1470 non-null  int64
 7   EducationField        1470 non-null  object
 8   EmployeeCount         1470 non-null  int64
 9   EmployeeNumber        1470 non-null  int64
10   EnvironmentSatisfaction 1470 non-null  int64
11   Gender               1470 non-null  object
12   HourlyRate           1470 non-null  int64
13   JobInvolvement        1470 non-null  int64
14   JobLevel             1470 non-null  int64
15   JobRole              1470 non-null  object
16   JobSatisfaction       1470 non-null  int64
17   MaritalStatus        1470 non-null  object
18   MonthlyIncome         1470 non-null  int64
19   MonthlyRate          1470 non-null  int64
20   NumCompaniesWorked    1470 non-null  int64
21   Over18               1470 non-null  object
22   OverTime             1470 non-null  object
23   PercentSalaryHike     1470 non-null  int64
24   PerformanceRating     1470 non-null  int64
25   RelationshipSatisfaction 1470 non-null  int64
26   StandardHours         1470 non-null  int64
27   StockOptionLevel     1470 non-null  int64
28   TotalWorkingYears     1470 non-null  int64
29   TrainingTimesLastYear 1470 non-null  int64
30   WorkLifeBalance       1470 non-null  int64
31   YearsAtCompany        1470 non-null  int64
32   YearsInCurrentRole    1470 non-null  int64
33   YearsSinceLastPromotion 1470 non-null  int64
34   YearsWithCurrManager  1470 non-null  int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	HourlyRate	JobInvolvement	JobLevel
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000
mean	36.923810	802.485714	9.192517	2.912925	1.0	1024.865306	2.721769	65.891156	2.729932	2.06
std	9.135373	403.509100	8.106864	1.024165	0.0	602.024335	1.093082	20.329428	0.711561	1.10
min	18.000000	102.000000	1.000000	1.000000	1.0	1.000000	1.000000	30.000000	1.000000	1.00
25%	30.000000	465.000000	2.000000	2.000000	1.0	491.250000	2.000000	48.000000	2.000000	1.00
50%	36.000000	802.000000	7.000000	3.000000	1.0	1020.500000	3.000000	66.000000	3.000000	2.00
75%	43.000000	1157.000000	14.000000	4.000000	1.0	1555.750000	4.000000	83.750000	3.000000	3.00
max	60.000000	1499.000000	29.000000	5.000000	1.0	2068.000000	4.000000	100.000000	4.000000	5.00

3 rows × 26 columns

df.isnull().sum()

```
Age                0
Attrition          0
BusinessTravel     0
DailyRate         0
Department        0
DistanceFromHome  0
Education          0
EducationField     0
EmployeeCount     0
EmployeeNumber     0
EnvironmentSatisfaction 0
Gender            0
HourlyRate        0
JobInvolvement    0
JobLevel          0
JobRole           0
JobSatisfaction   0
MaritalStatus     0
MonthlyIncome     0
MonthlyRate       0
NumCompaniesWorked 0
Over18            0
OverTime          0
PercentSalaryHike 0
PerformanceRating 0
RelationshipSatisfaction 0
StandardHours     0
StockOptionLevel  0
TotalWorkingYears 0
TrainingTimesLastYear 0
WorkLifeBalance   0
YearsAtCompany    0
YearsInCurrentRole 0
YearsSinceLastPromotion 0
YearsWithCurrManager 0
dtype: int64
```

for col in df.select_dtypes(include='object').columns:

```
df[col] = df[col].fillna(df[col].mode()[0])
```

for col in df.select_dtypes(include=np.number).columns:

```
df[col] = df[col].fillna(df[col].median())
```

```
df = df.drop(columns=["EmployeeNumber", "EmployeeCount", "Over18", "StandardHours"],
errors="ignore")
```

```
le = LabelEncoder()
```

for col in df.select_dtypes(include='object').columns:

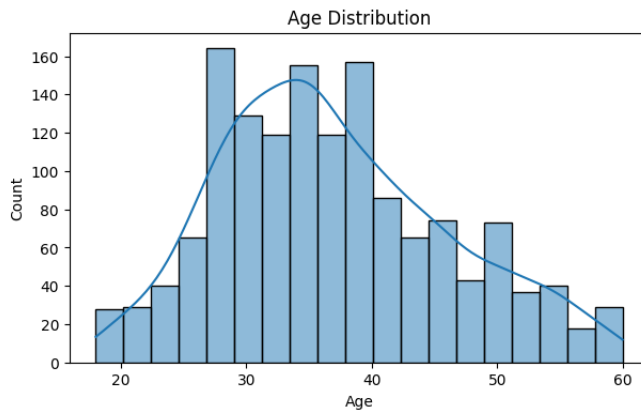
```
df[col] = le.fit_transform(df[col])
```

df.head()

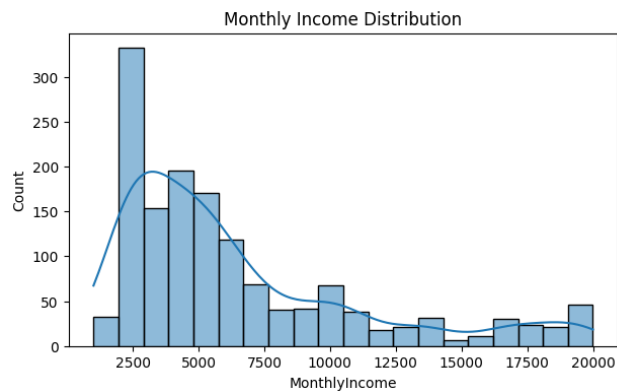
	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EnvironmentSatisfaction	Gender	...
0	41	1	2	1102	2	1	2	1	2	0	...
1	49	0	1	279	1	8	1	1	3	1	...
2	37	1	2	1373	1	2	2	4	4	1	...
3	33	0	1	1392	1	3	4	1	4	0	...
4	27	0	2	591	1	2	1	3	1	1	...

5 rows × 31 columns

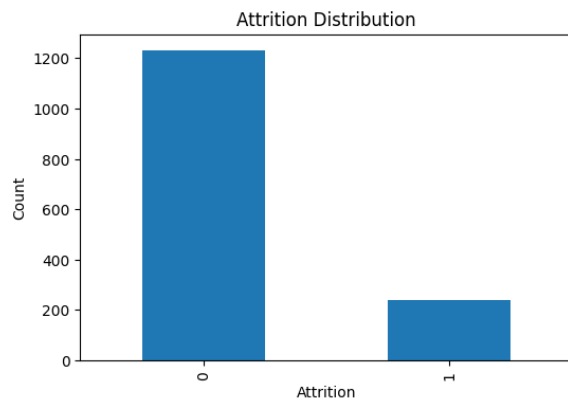
```
plt.figure(figsize=(7,4))
sns.histplot(df["Age"], kde=True)
plt.title("Age Distribution")
plt.show()
```



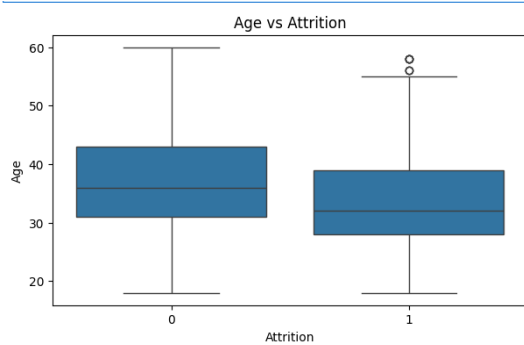
```
plt.figure(figsize=(7,4))
sns.histplot(df["MonthlyIncome"], kde=True)
plt.title("Monthly Income Distribution")
plt.show()
```



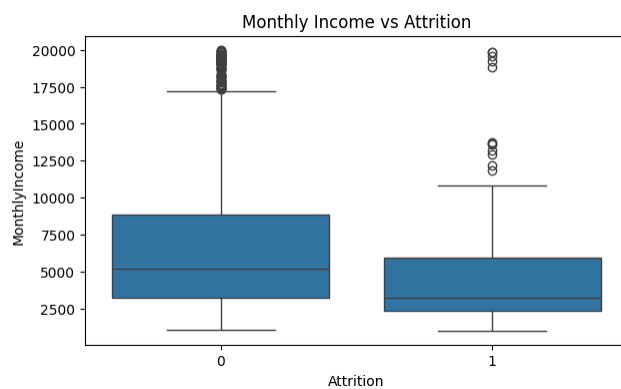
```
plt.figure(figsize=(6,4))
df["Attrition"].value_counts().plot(kind="bar")
plt.title("Attrition Distribution")
plt.xlabel("Attrition")
plt.ylabel("Count")
plt.show()
```



```
plt.figure(figsize=(7,4))  
sns.boxplot(x="Attrition", y="Age", data=df)  
plt.title("Age vs Attrition")  
plt.show()
```



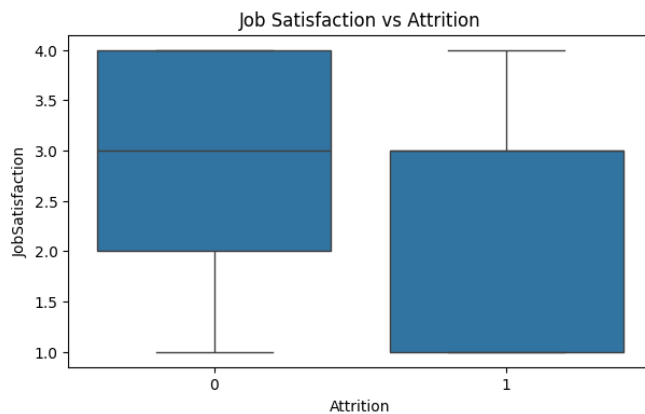
```
plt.figure(figsize=(7,4))  
sns.boxplot(x="Attrition", y="MonthlyIncome", data=df)  
plt.title("Monthly Income vs Attrition")  
plt.show()
```



```
plt.figure(figsize=(7,4))  
sns.boxplot(x="Attrition", y="JobSatisfaction", data=df)
```

```
plt.title("Job Satisfaction vs Attrition")
```

```
plt.show()
```

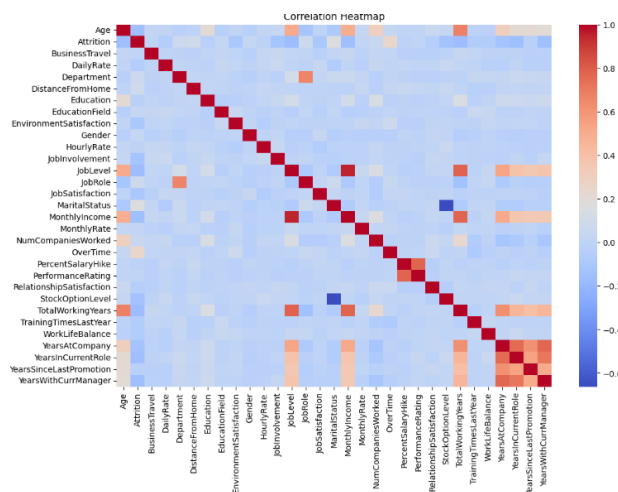


```
plt.figure(figsize=(12,8))
```

```
sns.heatmap(df.corr(),cmap="coolwarm")
```

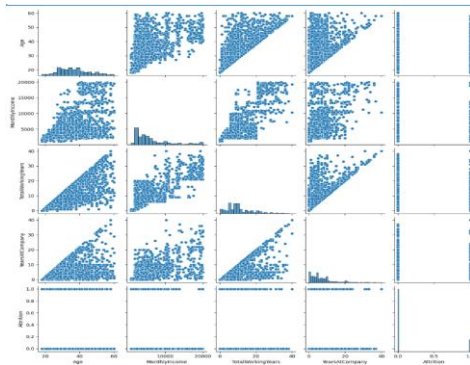
```
plt.title("Correlation Heatmap")
```

```
plt.show()
```

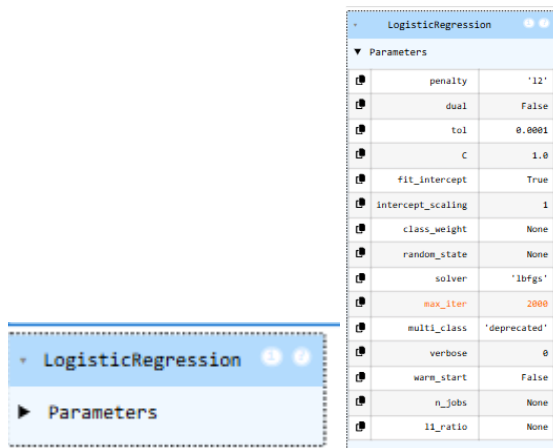


```
sns.pairplot(df[['Age','MonthlyIncome','TotalWorkingYears','YearsAtCompany','Attrition']])
```

```
plt.show()
```



```
X = df.drop("Attrition", axis=1)
y = df["Attrition"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = LogisticRegression(max_iter=2000)
model.fit(X_train, y_train)
```



```
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Accuracy: 0.8945578231292517